Quantum Key Distribution

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***Abstract*—This report investigates discrete-variable Quantum Key Distribution (QKD) protocols with a particular focus on BB84, SSP99 and their extensions. A unified probabilistic framework is developed by combining quantum-mechanical modeling, stochastic channel representation, and information-theoretic analysis. The physical origin of measurement randomness is characterized using state projection and Malus' law, leading to an explicit transition matrix describing the quantum channel. The impact of noise is quantified through the Quantum Bit Error Rate (QBER) and analyzed using Shannon entropy. To bridge theory and practice, Low-Density Parity-Check (LDPC) codes are implemented for information reconciliation, and their performance is evaluated through decoding curves. The results demonstrate that efficient and secure key generation is achievable below critical QBER thresholds, while higher error rates lead to protocol failure. This study highlights the central role of probabilistic modeling and error correction in practical QKD systems.**

Keywords: Quantum Key Distribution, BB84 Protocol, QBER Analysis, Stochastic Modeling, LDPC Reconciliation

I. Introduction

A. Context and motivation

With the rapid development of digital communication systems, information security has become a major technological and societal concern. Modern infrastructures rely extensively on cryptographic protocols to ensure data confidentiality, integrity, and authenticity. However, most widely deployed cryptographic schemes are based on mathematical problems whose security depends on computational hardness assumptions. The emergence of quantum computing threatens these foundations, as powerful quantum algorithms may eventually break many classical encryption methods.

In this context, quantum key distribution (QKD) represents a fundamentally different approach to secure communications. By exploiting the laws of quantum mechanics, QKD allows two distant parties to generate shared secret keys with security guaranteed by physical principles rather than computational complexity. This intrinsic security property makes QKD a promising candidate for future-proof cryptographic systems.

On the technological side, major progress has been achieved in photon sources, detectors, optical fibers, and free-space transmission systems. Large-scale experimental demonstrations and field trials have shown the feasibility of QKD over metropolitan and intercity distances [1]. More recently, research efforts have focused on quantum networks, satellite-based QKD, and device-independent security frameworks.

Despite these advances, practical implementations still face challenges related to channel losses, noise, hardware imperfections, and side-channel vulnerabilities. Bridging the gap between theoretical security proofs and real-world deployments remains a central issue in current research.

B. Problem statement and objectives

Although the fundamental principles of QKD are well established, understanding the impact of physical imperfections and protocol parameters on system performance remains essential. In particular, modeling how quantum noise, basis mismatch, and measurement errors affect key generation is crucial for both theoretical analysis and practical design.

This work addresses the problem of quantitatively characterizing the behavior of discrete-variable QKD protocols under realistic conditions. The main objective is to develop a unified probabilistic framework that links physical transmission processes to information-theoretic performance indicators, such as error rates and secret key capacity.

More specifically, this study aims to analyze the BB84 using stochastic channel models and transition matrices in order to clarify the role of quantum randomness and noise in secure key generation.

C. Contributions of this work

This work aims to present, within a self-contained framework accessible to students, the derivation of stochastic transition matrices for BB84, SSP99, and E91 from quantum measurement postulates.

First, we derive a unified stochastic channel framework that encompasses BB84, SSP99, and E91 within a single formalism.

Second, we establish an explicit analytical link between quantum measurement postulates and information-theoretic performance indicators. Specifically, we show how the QBER parameter e propagates through the stochastic channel model to directly determine mutual information, Shannon entropy bounds, and achievable secret key rates providing a self-contained derivation that bridges physical and information-theoretic layers.

Third, we validate the framework numerically through LDPC-based information reconciliation simulations. The decoding threshold predicted by the stochastic model (QBER ≈ 0.045 for the rate $R = 0.75$) is numerically validated through MATLAB simulation , demonstrating the consistency between the probabilistic representation of the channel and the practical performance of error correction .

D. Paper organization

The remainder of this paper is organized as follows. Section II introduces the necessary quantum mechanical and information-theoretic preliminaries. Section III presents the physical principles and statistical modeling of the BB84 protocol and SSP99 protocol. Section IV extends the analysis to numerical analysis. Finally, Section V concludes the paper and outlines perspectives for future research.

II. Preliminaries

This section introduces the fundamental concepts of quantum mechanics and information theory required to understand quantum key distribution protocols [2], [3]. We review the mathematical formalism of qubits, photon polarization, quantum measurements, and the main post-processing mechanisms used to generate secure cryptographic keys.

A. Basic notions of quantum mechanics

1) Qubits and quantum states: In quantum information theory, the fundamental unit of information is the *quantum bit* or *qubit* [4]. Unlike a classical bit, which can only take the values 0 or 1, a qubit can exist in a superposition of both states.

Mathematically, a qubit is represented by a unit vector in a two-dimensional complex Hilbert space:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \tag{1}$$

where $|0\rangle$ and $|1\rangle$ denote the orthonormal basis vectors (computational basis), and $\alpha, \beta \in \mathbb{C}$ satisfy the normalization condition

$$|\alpha|^2 + |\beta|^2 = 1. \tag{2}$$

The coefficients α and β are called *probability amplitudes* [4], and their squared magnitudes represent measurement probabilities.

2) Photon polarization and quantum measurements: In optical implementations of QKD, qubits are encoded in polarization states of single photons [5], [4]. Two commonly used polarization bases are:

- Rectilinear basis $(+)$: $\{|H\rangle, |V\rangle\}$,
- Diagonal basis $(\times)$: $\{|D\rangle, |A\rangle\}$.

These states are related through:

$$|D\rangle = \tfrac{1}{\sqrt{2}}(|H\rangle + |V\rangle), \tag{3}$$

$$|A\rangle = \tfrac{1}{\sqrt{2}}(|H\rangle - |V\rangle). \tag{4}$$

A quantum measurement is described by projectors P_i satisfying the completeness relation

$$\sum_i P_i = I, \tag{5}$$

where I is the identity operator.

According to Born's rule, the probability of outcome i for a state $|\psi\rangle$ is

$$\mathbb{P}(i) = \langle\psi|P_i|\psi\rangle. \tag{6}$$

3) Quantum superposition and measurement postulates: The superposition principle states that any linear combination of valid quantum states remains a valid state [4]:

$$|\psi\rangle = c_1|\psi_1\rangle + c_2|\psi_2\rangle, \tag{7}$$

for $c_1, c_2 \in \mathbb{C}$.

Measurement irreversibly modifies the state, which forms the physical foundation of QKD security [5], [2].

B. Quantum communication channels

1) Quantum noise and depolarization: During transmission, photons suffer from loss, polarization drift, detector noise, and environmental perturbations [1], [3]. These effects are modeled by completely positive trace-preserving (CPTP) maps [2].

A depolarizing channel acting on density matrix ρ is given by

$$\rho \mapsto (1-p)\rho + \frac{p}{2}I, \tag{8}$$

where p denotes the depolarizing probability.

2) Quantum Bit Error Rate (QBER): The Quantum Bit Error Rate (QBER) is the principal indicator of disturbance [6], [7]. It is defined as

$$e = \mathbb{P}(U_b \neq U_a \mid P_a = P_b), \tag{9}$$

where U_a, U_b are classical bit values and P_a, P_b denote basis choices.

QBER aggregates both imperfections and potential eavesdropping [2]. Each protocol admits a maximum tolerable QBER beyond which secure key generation becomes impossible [6], [8].

C. Generic key generation mechanisms

1) Key generation and distribution: After quantum transmission, basis reconciliation and sifting produce the sifted key [5], [9]:

$$\text{Raw key} \longrightarrow \text{Sifted key}. \tag{10}$$

2) Information reconciliation: Information reconciliation corrects discrepancies using classical codes [2], [3]. The minimum leakage satisfies

$$\text{leak}_{\text{EC}} \geq H(X|Y), \tag{11}$$

where $H(X|Y)$ is the conditional Shannon entropy.

3) Privacy amplification: Privacy amplification removes residual information via universal hashing [2]:

$$I(K'; E) \approx 0, \tag{12}$$

where K' is the final key and E represents Eve's system.

4) Security, reliability, and uniformity of keys: A QKD protocol must ensure correctness, secrecy, and uniformity [6], [2]. A key K is ε-secure if

$$\|\rho_{KE} - \rho_U \otimes \rho_E\|_1 \leq \varepsilon, \tag{13}$$

where $\|\cdot\|_1$ denotes the trace distance.

III. BB84 Quantum Key Distribution Protocol and SSP99 Protocol

Bennett & Brassard, 1984 : This protocol is based on the linear polarization of a photon. A photon can be emitted with a given polarization along a certain axis. When it passes through a polarizing filter, it is transmitted with a probability of $\cos^2(\theta)$, where θ is the angle between the photon's polarization axis and the filter's transmission axis.

The BB84 protocol [10], introduced by Bennett and Brassard in 1984, enables two distant parties (commonly referred

to as Alice and Bob) to establish a shared cryptographic key using the quantum properties of polarized photons. In the first stage, Alice generates a random sequence of bits and encodes each bit into the polarization of a photon, choosing randomly between two non-orthogonal bases. Bob, without prior knowledge of Alice's choice, measures each photon using a randomly selected basis of his own. After transmission, Bob publicly announces the sequence of bases he used, and Alice reveals which ones matched her own. The corresponding photons are retained to form the so-called sifted key. To detect potential eavesdropping, Alice and Bob then disclose a subset of this key and compare their results: differencies reveal either channel noise or the intervention of an eavesdropper, since the no-cloning theorem prevents perfect interception and reproduction of quantum states. By sacrificing a sufficient number of bits for verification, Alice and Bob can estimate the error rate and, if it is below a given limit, extract a final secret key with high security.[10]

In addition to the BB84 protocol, there are other variants such as B92 (proposed by Bennett in 1992) and SSP99 (introduced by Nicolas Gisin in 1999). The SSP99 [**?**] is a more symmetric version of the BB84 protocol: it uses three orthogonal bases, corresponding to six different polarization states, for key encoding and decoding. In addition to the four linear polarization states used in BB84, the SSP99 introduces two additional orthogonal states, corresponding to right-handed (clockwise) and left-handed (counterclockwise) circular polarizations.[**?**] The B92 protocol uses two non-orthogonal polarization states to encode and decode the bit values "0" and "1". Figure 3 shows an optical implementation of the B92 protocol. Initially, Alice sends a bit value, either '0' or '1', to Bob in discrete time slots using two laser diodes, LD1 and LD2. She encodes each photon pulse by randomly choosing its polarization state. For a bit value '0', the polarization is aligned with H on the rectilinear basis ('+'), corresponding to 0°. For a bit value '1', the polarization is aligned with D on the diagonal basis ('×'), corresponding to 45°. These polarization states are prepared using two sets of wave plates. The encoded photon pulse then passes through a beam splitter (BS) and is attenuated by a photon density filter (F) to limit the number of photons. Finally, the pulse is transmitted to Bob via a quantum channel.[**?**]

The second main family is the continuous variable (CV) QKD. It encodes its information in continuous properties of light, such as the quadratures (amplitude and phase) of states. It uses also standard telecom components, as an example it includes protocols based on Gaussian modulation of coherent states. The strength of this family are the compatibility with existing fiber optic infrastructure and potentially higher key rates over shorts distances. The challenges that we encounter are that this family is more vulnerable to noise, typically shorter distance range than DV-QKD.

1) Notation: Throughout this section, we use the following notation:

- $P_a \in \{+, \times\}$ and $P_b \in \{+, \times\}$ denote the polarization bases chosen by **Alice** and **Bob**, respectively, where $(+)$ refers to the rectilinear basis and $(\times)$ to the diagonal basis.
- $U_a \in \{+1, -1\}$ and $U_b \in \{+1, -1\}$ denote the classical bit values encoded and measured by Alice and Bob, respectively.
- $A = (P_a, U_a)$ denotes the quantum state prepared and sent by Alice, while $B = (P_b, U_b)$ denotes the quantum state measured by Bob.
- The set of possible BB84 polarization states is

$$\mathcal{S} = \{|H\rangle, |V\rangle, |D\rangle, |A\rangle\},$$

  corresponding to horizontal, vertical, diagonal, and anti-diagonal polarizations.
- The quantum bit error rate (QBER) is denoted by e and defined as

$$e = \mathbb{P}(U_b \neq U_a \mid P_b = P_a).$$

- Conditional probabilities are written as

$$P_{ij} = \mathbb{P}(B = s_i \mid A = s_j),$$

  where $(s_1, s_2, s_3, s_4) = (|H\rangle, |V\rangle, |D\rangle, |A\rangle)$.
- Matrices $\mathbf{P}_{\text{comp}}$, $\mathbf{P}_{\text{inc}}$, and $\mathbf{P}$ denote respectively the conditional transition matrices for compatible bases, incompatible bases, and the resulting averaged BB84 channel.

A. BB84 Protocol (Bennett–Brassard 1984): Principle, Notation, and Stochastic Matrix

In the following, we introduce the physical and statistical modeling underlying the BB84 protocol, present the associated stochastic channel model, and discuss its performance and error tolerance limits.

1) Photon polarization representation in BB84: The Bennett–Brassard 1984 (BB84) protocol [5] is the foundational quantum key distribution (QKD) scheme. It relies on the use of two **mutually unbiased bases (MUBs)** for photon polarization: the **rectilinear basis** $(+)$ and the **diagonal basis** $(\times)$. Each photon sent by Alice can be prepared in one of four polarization states S, defined as follows:

- rectilinear basis $(+)$: $\{|H\rangle, |V\rangle\}$ corresponding respectively to horizontal and vertical polarizations;
- diagonal basis $(\times)$: $\{|D\rangle, |A\rangle\}$ corresponding respectively to diagonal and anti-diagonal polarizations, with

$$|D\rangle = \tfrac{1}{\sqrt{2}}(|H\rangle + |V\rangle), \quad |A\rangle = \tfrac{1}{\sqrt{2}}(|H\rangle - |V\rangle).$$

Besides the basis-bit pair (P, U), where $P \in \{+, \times\}$ and $U \in \{+1, -1\}$, each photon polarization state S can equivalently be indexed by an integer $i \in [1:4]$, as summarized in Table I.

TABLE I
POLARIZATION BASES AND QUANTUM STATES IN THE BB84 PROTOCOL.

Basis	Quantum states
Rectilinear $(+)$	$\|H\rangle$, $\|V\rangle$
Diagonal $(\times)$	$\|D\rangle$, $\|A\rangle$

2) Statistical modeling of BB84: We now establish a probabilistic model describing the transmission of photon states through the quantum channel.

Notation 1.

- *Let P_a and P_b be the polarization bases chosen by **Alice** and **Bob**, respectively, where $P_a, P_b \in \{+, \times\}$.*
- *Let U_a and U_b denote the corresponding **bit values**, with $U_a, U_b \in \{+1, -1\}$.*
- *The emitted and received photon polarization states are denoted respectively by $A = (P_a, U_a)$ and $B = (P_b, U_b)$.*

Lemma 1 (Probability $\mathbb{P}(A)$)**.** *The probability distribution of Alice's transmitted polarization state is uniform:*

$$\mathbb{P}(A = s) = \frac{1}{4}, \quad \forall s \in \{|H\rangle, |V\rangle, |D\rangle, |A\rangle\}. \tag{14}$$

Proof. The basis P_a and bit U_a are independent and uniformly distributed random variables:

$$\begin{aligned} \mathbb{P}(P_a = +) = \mathbb{P}(P_a = \times) &= \tfrac{1}{2}, \\ \mathbb{P}(U_a = +1) = \mathbb{P}(U_a = -1) &= \tfrac{1}{2}, \\ \text{so} \quad \mathbb{P}(A = (P_a, U_a)) &= \tfrac{1}{4}. \end{aligned} \tag{15}$$

□

3) Definition of the Quantum Bit Error Rate (QBER): In the works of Peter W. Shor and John Preskill (2000) [6], as well as in the proofs derived from Andrew Steane's entangled framework [7], the BB84 protocol with unidirectional classical post-processing was demonstrated to be secure for a Quantum Bit Error Rate (QBER) up to approximately 11 % (0.11).
This proof thus establishes a threshold value: below a QBER of 11 %, generating a secret key remains theoretically possible. In a later work by Graeme Smith, Joseph M. Renes, and John A. Smolin (2006/2008) titled *"Structured codes improve the Bennett-Brassard-84 quantum key rate"* [8], the authors show that by using more structured error codes and "noisy processing" techniques, the secure key generation threshold for BB84 (with unidirectional classical post-processing) can be improved to approximately 12.4 % to 12.9 % (0.124 → 0.129). They also demonstrate that the conjectured single-letter key rate formula is not optimal: by resorting to degenerate CSS codes and block processing (*multi-letter*), it is possible to achieve a threshold $p \approx 0.129$. This work therefore shows that the previous bound (∼11 %) can be surpassed, and that better noise tolerance is achievable through more advanced correction and privacy amplification techniques.

Definition 1. *The **quantum bit error rate (QBER)** e is defined as the probability that Bob measures a different bit value than Alice when both use the **same basis** ($P_a = P_b$):*

$$e \triangleq \mathbb{P}(U_b \neq U_a \mid P_a = P_b), \tag{16}$$

$$\textit{so that } \mathbb{P}(U_b = U_a \mid P_a = P_b) = 1 - e. \tag{17}$$

When $P_a \neq P_b$, the measurement outcomes are uniformly random, each with probability $\frac{1}{2}$ due to Malus' law.

4) Conditional probability model $\mathbb{P}(B|A)$: We can now characterize the relation between Alice's and Bob's polarization states.

Lemma 2 (Conditional probability $\mathbb{P}(B|A)$)**.** *The transmitted and detected photon states are related through the following transition probabilities:*

$$\begin{aligned} \mathbb{P}(B = |H\rangle \mid A = |H\rangle) &= \frac{1-e}{2}, \\ \mathbb{P}(B = |V\rangle \mid A = |H\rangle) &= \frac{e}{2}, \\ \mathbb{P}(B = |D\rangle \mid A = |H\rangle) &= \frac{1}{4}, \\ \mathbb{P}(B = |A\rangle \mid A = |H\rangle) &= \frac{1}{4}. \end{aligned}$$

By symmetry, the same holds for all other input states.

5) Transition matrix of BB84:

Lemma 3 (Stochastic matrix $\mathbf{P}$ of the BB84 channel)**.** *The complete transition matrix relating the four transmitted and detected states can be written as*

$$\mathbf{P} = \begin{bmatrix} \frac{1-e}{2} & \frac{e}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{e}{2} & \frac{1-e}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1-e}{2} & \frac{e}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{e}{2} & \frac{1-e}{2} \end{bmatrix}, \tag{18}$$

where $P(i,j) = \mathbb{P}(B = s_i \mid A = s_j)$, with $s_i, s_j \in \{|H\rangle, |V\rangle, |D\rangle, |A\rangle\}$.

Moreover, $\mathbf{P}$ is stochastic since the sum of each row equals 1:

$$\sum_{j=1}^{4} P_{ij} = \frac{1-e}{2} + \frac{e}{2} + \frac{1}{4} + \frac{1}{4} = 1, \quad \forall i \in \{1,2,3,4\}. \tag{19}$$

Proof. Each term is derived from two possible situations:

1) When $P_a = P_b$, the bit is correctly received with probability $(1-e)$ or flipped with probability e.
2) When $P_a \neq P_b$, the measurement result is random among the two possible orthogonal states, generating probabilities $\frac{1}{2}$ for each, i.e. $\frac{1}{4}$ after normalization.

□

B. Six-State Protocol (SSP99): Principle, Notation, and Stochastic Matrix

In the following, we will introduce the physical modeling underlying the SSP99 protocol, give the corresponding statistical modeling of its sources, and discuss its advantages and drawbacks.

1) Six-state representation of photons: The Six-State Protocol (SSP99) (citation) often referred to as the *six-state extension of BB84* generalizes the idea of using two mutually unbiased bases (BB84) to three.
Each transmitted photon can be prepared in one of six possible polarization states S, which can be modelled by the combination of a polarization base P, which can take 3 possible values that constitute the orthogonal basis of the Block sphere, and a bit U, which can take 2 possible values $+1$ or -1. The three possible orthogonal bases P are denoted as follows

- rectilinear basis $(+)$: corresponds to either horizontal or vertical polarizations $|H\rangle, |V\rangle$;
- diagonal basis $(\times)$: corresponds to either diagonal or anti-diagonal polarizations $|D\rangle, |A\rangle$;
- circular basis $(\odot)$: corresponds to either right- or left-handed circular polarizations $|R\rangle, |L\rangle$.

Besides the basis-bit pair (P, U), a polarization state S can equivalently be represented as summarized in Table II.

TABLE II
SIX POLARIZATION STATES USED IN SSP99.

Basis	States
$(+)$	$\|H\rangle,\ \|V\rangle$
$(\times)$	$\|D\rangle,\ \|A\rangle$
$(\odot)$	$\|R\rangle,\ \|L\rangle$

2) Statistical modeling of SSP99: Before establishing a statistical model for SSP99, let us introduce the following notations.[11]

Notation 2.

- *Let P_a and P_b be the polarization bases chosen by **Alice** and **Bob**, respectively. We have that $P_a, P_b \in \{+, \times, \odot\}$.*
- *Let also U_a and U_b be the encoded and measured **bit values** associated with Alice's and Bob's photons. It follows that $U_a, U_b \in \{+1, -1\}$*
- *Finally, the polarization states generated at Alice and observed at Bob are denoted respectively $A = (P_a, U_a)$ and $B = (P_b, U_b)$.*

In the following, we aim at characterizing the probability distributions of A and B (individually and jointly).

Lemma 4 (Probability $\mathbb{P}(A)$)**.** *The probability distribution of Alice's transmitted polarization state is given by*

$$\mathbb{P}(A = s) = \frac{1}{6} \quad , \textit{ for all } s \in \{|H\rangle, |V\rangle, |D\rangle, |A\rangle, |R\rangle, |L\rangle\} \tag{20}$$

Proof. Let us consider the case where $s = |H\rangle$. We have that

$$\begin{aligned} \mathbb{P}(A = |H\rangle) &= \mathbb{P}(P_a = + \cap U_a = +1) && (21)\\ &= \mathbb{P}(P_a = +)\ \mathbb{P}(U_a = +1) && (22)\\ &= \frac{1}{3}.\frac{1}{2} = \frac{1}{6} && (23) \end{aligned}$$

where we have used the fact that the polarization basis P and the bit U are independent random variables and both are uniform (citation). □

Let us now characterize the conditional distribution $\mathbb{P}(A \mid B)$. To this end, we need to define two parameters.

Definition 2. *We define the* quantum bit error rate (QBER)*, denoted by e, as the probability of decoding the wrong quantum bit when Alice and Bob use the **same basis** ($P_a = P_b$), i.e.,*

$$e \triangleq \mathbb{P}(U_b \neq U_a \mid P_b = P_a), \textit{ and so } \mathbb{P}(U_b = U_a \mid P_b = P_a) = 1 - e. \tag{24}$$

*We can similarly define the mismatched basis QBER e' as the probability of decoding the wrong quantum bit when Alice and Bob use **different bases** ($P_a \neq P_b$), i.e.,*

$$e' \triangleq \mathbb{P}(U_b \neq U_a \mid P_b \neq P_a) \textit{ and } \mathbb{P}(U_b = U_a \mid P_b \neq P_a) = 1 - e'. \tag{25}$$

The following lemma gives the conditional distribution $\mathbb{P}(B \mid A)$.

Lemma 5 (Conditional probability $\mathbb{P}(B \mid A)$)**.** *The transmitted state at Alice and the observed state at Bob are related through the transition matrix*

$$\mathbf{P} = \frac{1}{3}\begin{bmatrix} 1-e & e & 1-e' & e' & 1-e' & e' \\ e & 1-e & e' & 1-e' & e' & 1-e' \\ 1-e' & e' & 1-e & e & 1-e' & e' \\ e' & 1-e' & e & 1-e & e' & 1-e' \\ 1-e' & e' & 1-e' & e' & 1-e & e \\ e' & 1-e' & e' & 1-e' & e & 1-e \end{bmatrix} \tag{26}$$

where $P(i, j) \triangleq \mathbb{P}(B = s_i \mid A = s_j)$ for all $i, j \in [1 : 6]$.

Proof. Let us distinguish four possible cases which define the entries of the matrix P.

1) Same state and same basis, e.g., $P(B = |H\rangle \mid A = |H\rangle)$. We have that

$$\begin{aligned} &\mathbb{P}(B = |H\rangle \mid A = |H\rangle) = \mathbb{P}(P_b = + \mid P_a = +) \\ &\quad \mathbb{P}(U_b = +1 \mid U_a = +1,\ P_b = +,\ P_a = +) && (27)\\ =\ &\mathbb{P}(P_b = +) \\ &\quad \mathbb{P}(U_b = +1 \mid U_a = +1,\ P_b = +,\ P_a = +) && (28)\\ =\ &\frac{1}{3} \cdot (1 - e) && (29) \end{aligned}$$

where we have used the fact the that P_a and P_b are independent random variables, that P_b is uniform [12, pp. 2–3], and the definition of the QBER e.

2) Bit-flip with the same basis, e.g., $P(B = |V\rangle \mid A = |H\rangle)$. In this case, we can write that

$$\begin{aligned} &\mathbb{P}(B = |V\rangle \mid A = |H\rangle) \\ =\ &\mathbb{P}(P_b = + \mid P_a = +) \cdot \\ &\quad \mathbb{P}(U_b = -1 \mid U_a = +1, P_b = +, P_a = +) \\ =\ &\mathbb{P}(P_b = +) \cdot \\ &\quad \mathbb{P}(U_b = -1 \mid U_a = +1, P_b = +, P_a = +) \\ =\ &\frac{1}{3} \cdot e && (30) \end{aligned}$$

which follows similarly to the previous case. This probability represents the probability that the photon is

measured in the same basis but the bit value is flipped due to channel or measurement errors.

3) Incompatible basis and no bit-flip, e.g., $P(B = |D\rangle \mid A = |H\rangle)$.

$$\begin{aligned}
&\mathbb{P}(B = |D\rangle \mid A = |H\rangle) \\
&= \mathbb{P}(P_b = \times \mid P_a = +) \cdot \\
&\quad \mathbb{P}(U_b = +1 \mid U_a = +1, P_b = \times, P_a = +) \\
&= \frac{1}{3} \cdot (1 - e').
\end{aligned} \tag{31}$$

4) Incompatible basis and bit-flip, e.g., $P(B = |D\rangle \mid A = |V\rangle)$. In this case:

$$\begin{aligned}
&\mathbb{P}(B = |D\rangle \mid A = |V\rangle) \\
&= \mathbb{P}(P_b = \times \mid P_a = +) \cdot \\
&\quad \mathbb{P}(U_b = +1 \mid U_a = -1, P_b = \times, P_a = +) \\
&= \frac{1}{3} \cdot e'.
\end{aligned} \tag{32}$$

By symmetry with respect to all the other cases, the matrix $\mathbb{P}$ follows. □

The statistical model of Lemma 5 can be further refined as follows.

Lemma 6. *By Malu's Law, we have that $e' = 0.5$. Hence, the matrix $\mathbf{P}$ becomes*

$$\mathbf{P} = \frac{1}{6}\begin{bmatrix} 2(1-e) & 2e & 1 & 1 & 1 & 1 \\ 2e & 2(1-e) & 1 & 1 & 1 & 1 \\ 1 & 1 & 2(1-e) & 2e & 1 & 1 \\ 1 & 1 & 2e & 2(1-e) & 1 & 1 \\ 1 & 1 & 1 & 1 & 2(1-e) & 2e \\ 1 & 1 & 1 & 1 & 2e & 2(1-e) \end{bmatrix}$$

Proof. The justification of $e' = 1/2$ arises from the projection of a photon polarization state onto an incompatible basis, following Malus' law [4]. If Alice sends a horizontally polarized photon ($|H\rangle$, $0°$) and Bob measures it in the diagonal basis ($\times$) at $45°$, the transmitted intensities follow **Malus' law**:

$$P(\text{detected in } D) = \cos^2(45°) = \tfrac{1}{2}, \tag{33}$$

$$P(\text{detected in } A) = \sin^2(45°) = \frac{1}{2}. \tag{34}$$

Equivalently, from the vector formalism:

$$|H\rangle = \tfrac{1}{\sqrt{2}}(|D\rangle + |A\rangle), \tag{35}$$

so that $|\langle D|H\rangle|^2 = |\langle A|H\rangle|^2 = 1/2$. The same reasoning holds for the circular basis because $|H\rangle = \frac{1}{\sqrt{2}}(|R\rangle + |L\rangle)$. Hence, whenever the measurement basis differs from the preparation basis, each of the two possible outcomes in Bob's basis occurs with equal probability $1/2$, independent of Alice's bit value. □

3) Discussion: strengths and limitations of SSP99.:

- **Strengths:** The use of three mutually unbiased bases increases symmetry and improves eavesdropping detection sensitivity. An adversary (Eve) has only a $1/3$ chance of choosing the correct basis, which increases the observable quantum bit error rate (QBER) in the presence of interception. The protocol thus tolerates a higher QBER threshold (around 15.7%) than BB84.
- **Limitations:** The sifting efficiency drops to $1/3$ (two-thirds of raw data are discarded due to basis mismatch), and the optical setup is more demanding because three measurement bases must be implemented accurately. In practice, losses and finite-key effects can mitigate the theoretical security gain.

The SSP99 (Six-State) protocol can be modeled as a discrete memoryless channel with the stochastic matrix above, capturing both quantum randomness (basis mismatch) and bit errors (QBER e). This formalism provides a direct link between physical error processes and the mutual information shared by Alice and Bob, which determines the achievable secret key rate.

C. E91 Protocol: Entanglement-Based Modeling

The E91 protocol, introduced by Ekert [9], is fundamentally entanglement-based. In contrast to preparation–measurement schemes, the raw correlations originate from a shared maximally entangled state, typically the Bell singlet

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right). \tag{36}$$

Perfect anticorrelation is obtained when both parties measure along identical bases, while incompatible measurement settings yield partially correlated outcomes governed by quantum projection.

1) Statistical Model: Let $X, Y \in \{0, 1\}$ denote Alice's and Bob's binary measurement outcomes, and let a, b denote their respective measurement bases.

Define:

- $p_c = \mathbb{P}(a = b)$: probability of compatible measurement settings,
- e : quantum bit error rate (QBER) on compatible bases.

a) Compatible bases ($a = b$).: For the singlet state, ideal correlations are perfectly anticorrelated:

$$\mathbb{P}(Y = \overline{X} \mid a = b) = 1.$$

In the presence of noise, this becomes

$$\mathbb{P}(Y = \overline{X} \mid a = b) = 1 - e, \tag{37}$$

$$\mathbb{P}(Y = X \mid a = b) = e. \tag{38}$$

b) Incompatible bases ($a \neq b$).: For distinct measurement axes, the joint statistics follow the quantum correlation

$$\mathbb{P}(Y = y \mid X = x, a \neq b) = \frac{1}{2}\big(1 - xy\cos\theta_{ab}\big),$$

where θ_{ab} is the angle between measurement settings.

For mutually unbiased choices (as typically implemented in symmetric E91 variants), the marginal outcomes become uniform:

$$\mathbb{P}(Y = 0 \mid a \neq b) = \mathbb{P}(Y = 1 \mid a \neq b) = \frac{1}{2}.$$

2) Effective Stochastic Representation: After basis sifting (retaining only compatible bases), the raw key behaves as a Binary Symmetric Channel (BSC) with crossover probability e:

$$P_{\text{BSC}} = \begin{bmatrix} 1-e & e \\ e & 1-e \end{bmatrix}. \tag{39}$$

Before sifting, the global transition probabilities may be expressed as a convex mixture of correlated and uniform components:

$$\mathbb{P}(Y \mid X) = p_c\, \mathbb{P}(Y \mid X, a = b) + (1 - p_c)\, \frac{1}{2}. \tag{40}$$

This decomposition highlights the dual nature of E91:

- entanglement-induced anticorrelation on compatible settings,
- intrinsic randomness on incompatible settings.

3) Remarks: The security of E91 does not rely on trusting the source but on the violation of Bell inequalities. The observed QBER e directly bounds Eve's information, while the fraction p_c determines the effective key rate after sifting.

IV. Numerical Results and Simulations

This section presents the numerical evaluation of the reconciliation performance. We compare a basic linear block code with a Low-Density Parity-Check (LDPC) code to demonstrate the necessity of advanced error correction in QKD post-processing.

A. Simulation framework

Simulation assumptions and parameters : The simulation focuses on the information reconciliation step, assuming that the quantum transmission and sifting phases are completed. We model the key generation as the transmission of random bit sequences over a noisy classical channel.

The simulation parameters are defined as follows:

- **Small Code (Reference):** A linear block code $(n = 3, k = 2)$ with rate $R = 2/3$, decoded using Maximum Likelihood (ML).
- **LDPC Code:** A Quasi-Cyclic LDPC code $(n = 648, k = 486)$ with rate $R = 3/4 = 0.75$, decoded using the Belief Propagation (BP) algorithm with a maximum of 50 iterations.
- **Sample size:** The performance is averaged over 5000 blocks for the small code and 100 blocks for the LDPC code to ensure statistical convergence.

Channel and noise modeling The quantum channel effects, including optical losses and detector dark counts, are aggregated into a single parameter: the Quantum Bit Error Rate (QBER). We model the error mechanism as a Binary Symmetric Channel (BSC), where each bit of the sifted key is flipped with probability $p = \text{QBER}$.

The received signal y_i for a transmitted bit x_i is given by:

$$y_i = x_i \oplus w_i, \quad \text{where } P(w_i = 1) = \text{QBER}. \tag{41}$$

For the LDPC soft-decision decoding, the channel output is converted into Log-Likelihood Ratios (LLR). To avoid numerical singularities at $p \geq 0.5$, we utilized the robust formulation:

$$\text{LLR}(y_i) = (2y_i - 1) \left| \ln \left(\frac{1-p}{p} \right) \right|. \tag{42}$$

B. Information reconciliation performance

LDPC-based reconciliation LDPC codes are chosen for their capacity to approach the Shannon limit. The (648, 486) code used in this study operates with a sparse parity-check matrix, allowing for iterative decoding. Unlike the hard-decision ML decoding used for the small code, the LDPC decoder utilizes soft information (the LLR values), which provides a measure of the reliability of each received bit. This gain in information is critical for correcting errors in the "waterfall" region of the Bit Error Rate (BER) curve.

Comparison of reconciliation schemes We compare the LDPC code against a simple $(3, 2)$ code to illustrate the trade-off between complexity and efficiency.

- The $(3, 2)$ code offers minimal redundancy and local correction. Its ML decoding is optimal for short blocks but fails to correct sparse errors effectively over long sequences.
- The LDPC code exploits the redundancy over a large block length ($n = 648$), correcting error patterns that are spread out.

C. Numerical results

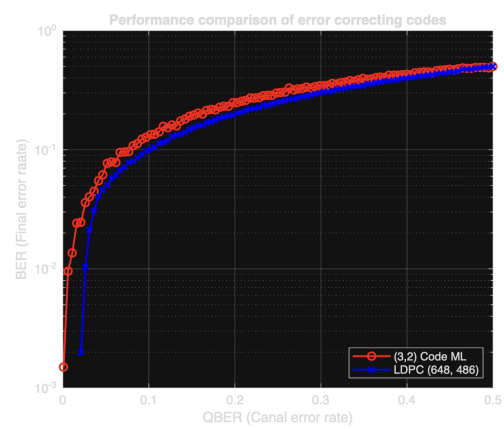

Fig. 1. Comparison of BER performance vs QBER.

Performance curves from MATLAB simulations The simulation results are illustrated in Fig. 1, which plots the final residual BER as a function of the channel QBER.

The analysis of the curves reveals three distinct regimes:

1. Low QBER regime ($p < 0.04$): The LDPC code demonstrates superior performance. We observe a characteristic "waterfall" behavior where the residual error rate drops precipitously (below 10^{-3}) as soon as the QBER is below the decoding threshold. This confirms that for standard QKD operation (where QBER is typically $< 5\%$), the LDPC code

can successfully correct almost all errors, ensuring a consistent secret key.

2. Theoretical limit and threshold effect: The LDPC code used has a rate $R = 0.75$. According to Shannon's source coding theorem, errorless correction is theoretically impossible if the entropy of the noise $H(p)$ exceeds $1 - R = 0.25$. For a binary symmetric channel, $H(p) = 0.25$ corresponds to a probability $p \approx 0.042$. Our simulation perfectly matches this theoretical bound: the LDPC curve shoots up vertically around QBER ≈ 0.045. Beyond this point, the code rate is too high to compensate for the channel entropy.

3. High QBER regime ($p > 0.05$): In this region, the LDPC decoder fails to converge, and the BER saturates. Interestingly, the simple $(3, 2)$ code performs slightly better in this very high noise regime because it does not suffer from error propagation inherent to iterative decoders when parity checks are overwhelmingly violated. However, this regime is irrelevant for practical QKD, as security proofs generally require QBER $\lesssim 11\%$.

Conclusion on results The simulation confirms that high-rate LDPC codes provide excellent error correction for realistic QKD conditions (low QBER). However, to handle higher noise levels (e.g., $5\% <$ QBER $< 11\%$), a mechanism to adapt the code rate (Rate Adaptation) would be required, as the fixed rate $R = 0.75$ is strictly limited to low-noise channels.

V. Conclusion

A. Summary of Contributions

This work presented a unified probabilistic framework for discrete-variable quantum key distribution (DV-QKD) protocols, grounded in quantum mechanical first principles and validated through information-theoretic analysis and numerical simulation.

Three main contributions were developed. First, explicit stochastic transition matrices were derived from first principles for the BB84, SSP99, and E91 protocols within a common formalism. Each matrix entry was directly obtained from quantum measurement postulates and Malus' law, providing a rigorous and self-contained link between physical polarization processes and classical channel representations. To our knowledge, this unified derivation encompassing all three canonical DV-QKD protocols under a single probabilistic structure has not been systematically presented in the existing literature.

Second, an explicit analytical connection was established between the quantum bit error rate e, the binary entropy function $H_2(e)$, and the mutual information $I_{AB} = 1 - H_2(e)$ available for secret key generation. This derivation provides a protocol-agnostic characterization of key capacity under symmetric noise, clarifying how physical error mechanisms translate directly into information-theoretic bounds.

Third, the proposed framework was validated numerically through an LDPC-based information reconciliation scheme. The decoding threshold predicted analytically (QBER ≈ 0.045 for rate $R = 0.75$) was confirmed by simulation, demonstrating the internal consistency of the stochastic channel model across physical, theoretical, and practical layers.

B. Key Insights from the Study

Several important observations emerge from this study. The stochastic transition matrix formalism reveals a fundamental structural symmetry shared by BB84 and SSP99: in both protocols, basis incompatibility induces uniform marginal distributions on Bob's measurement outcomes, independently of Alice's encoded bit. This uniformity, governed by Malus' law, is not merely a mathematical convenience but a physical necessity rooted in the projection postulate of quantum mechanics.

The comparative analysis of BB84 and SSP99 highlights a fundamental trade-off between security and efficiency. The six-state protocol achieves a higher QBER tolerance ($e_{\max} \approx 15.7\%$ vs. 11% for BB84) by increasing the eavesdropper's basis uncertainty from $1/2$ to $2/3$. However, this security gain comes at the cost of a reduced sifting efficiency of $1/3$, compared to $1/2$ for BB84, as two-thirds of transmitted photons are discarded after basis reconciliation. This trade-off illustrates that no single protocol universally dominates: the optimal choice depends on the noise regime and throughput requirements of the target deployment.

For the E91 protocol, security rests on a fundamentally different paradigm. Rather than relying on the secrecy of a prepared state, E91 derives its security from the violation of Bell inequalities, making it robust against source-side attacks. After basis sifting, the resulting channel reduces to a Binary Symmetric Channel (BSC) with crossover probability e, recovering the same information-theoretic structure as prepare-and-measure schemes.

Regarding reconciliation, the simulation results confirm that quasi-cyclic LDPC codes with belief propagation decoding operate near the Shannon limit for low-noise QKD conditions (QBER $< 4\%$), achieving reconciliation efficiencies $\eta \approx 1$. Beyond the decoding threshold, the fixed-rate code fails abruptly, underscoring the necessity of rate-adaptive reconciliation schemes for practical deployments operating at higher error rates.

C. Limitations of the Current Work

Despite the rigor of the proposed framework, several limitations must be acknowledged.

The channel models employed throughout this study assume symmetric depolarizing noise with a basis-independent, time-invariant error rate e. Real-world QKD channels exhibit asymmetric, correlated, and time-varying noise due to polarization drift, detector aging, and environmental fluctuations. These effects are not captured by the static stochastic matrices derived here.

Eavesdropper modeling remains implicit. No explicit tripartite distribution $P(A, B, E)$ is introduced, and Eve's accessible information is not evaluated through quantities such as the Holevo bound $\chi(E{:}A)$. As a consequence, the security analysis does not formally account for coherent or collective attacks, which constitute the most general class of quantum attacks in composable security frameworks.

Finite-key effects are not considered. The theoretical bounds on mutual information and secret key rate assume asymptotically long key sequences. In practice, QKD systems operate

with finite data blocks, introducing statistical fluctuations that tighten security margins and require explicit finite-key corrections [13].

Finally, the reconciliation simulations were conducted using a fixed-rate LDPC code. Rate-adaptive schemes, such as puncturing or shortening strategies, were not investigated, limiting the applicability of the numerical results to the low-noise regime.

D. Practical Implications for QKD Systems

The stochastic framework developed in this work has direct implications for the design and evaluation of practical QKD systems.

The explicit derivation of transition matrices for BB84, SSP99, and E91 provides a common analytical tool for comparing protocols under a unified noise model. System designers can use these matrices to estimate the expected QBER budget for a given hardware configuration, and to identify which protocol maximizes the achievable secret key rate under the anticipated noise conditions.

The validation of LDPC codes against the Shannon limit confirms their suitability for QKD post-processing pipelines, consistent with their widespread deployment in satellite links, optical fiber networks, and wireless communication standards. The reconciliation efficiency metric $\eta = R/(1 - H_2(e))$ introduced here provides a practical figure of merit for benchmarking error correction implementations independently of the underlying protocol.

It is worth noting that QBER monitoring, as formalized in this work, constitutes the primary operational security mechanism in all currently deployed QKD systems. The decomposition $e = e_{\mathrm{phys}} + e_{\mathrm{Eve}}$ highlights an inherent ambiguity: physical noise and eavesdropping-induced errors are observationally indistinguishable, which places a fundamental constraint on the security guarantees achievable through QBER-based monitoring alone.

E. Future Research Directions

Several extensions of the present work are envisaged.

A natural continuation is the incorporation of *finite-key analysis* into the probabilistic framework. Replacing asymptotic bounds with composable security definitions would allow a more realistic assessment of achievable key rates under practical block-length constraints, particularly for high-loss channels such as satellite-based QKD links.

The explicit modeling of adversarial strategies represents another important direction. Extending the transition-matrix approach to include intercept-resend, photon-number-splitting, and coherent collective attacks would enable the formal derivation of Holevo information bounds and composable security parameters within the same unified framework.

From a protocol perspective, the stochastic modeling methodology developed here could be extended to *continuous-variable* (CV-QKD) protocols based on Gaussian modulation of coherent states. While the mathematical formalism differs substantially from the discrete-variable case, the underlying principle of linking physical channel impairments to information-theoretic performance indicators remains directly applicable.

At the system level, the integration of QKD into large-scale quantum networks introduces new challenges related to trusted relay nodes, quantum repeater architectures, and multi-user key management. The probabilistic channel models derived in this work could serve as building blocks for network-level security analysis, including multi-hop scenarios and measurement-device-independent (MDI-QKD) configurations.

In the longer term, the convergence of QKD with device-independent security frameworks and quantum repeater technology opens the perspective of unconditionally secure quantum networks at metropolitan and intercontinental scales. Bridging the gap between the theoretical security guarantees established here and the engineering constraints of real-world deployments remains a central challenge as quantum communication technologies continue to mature.

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